

IT Project Management

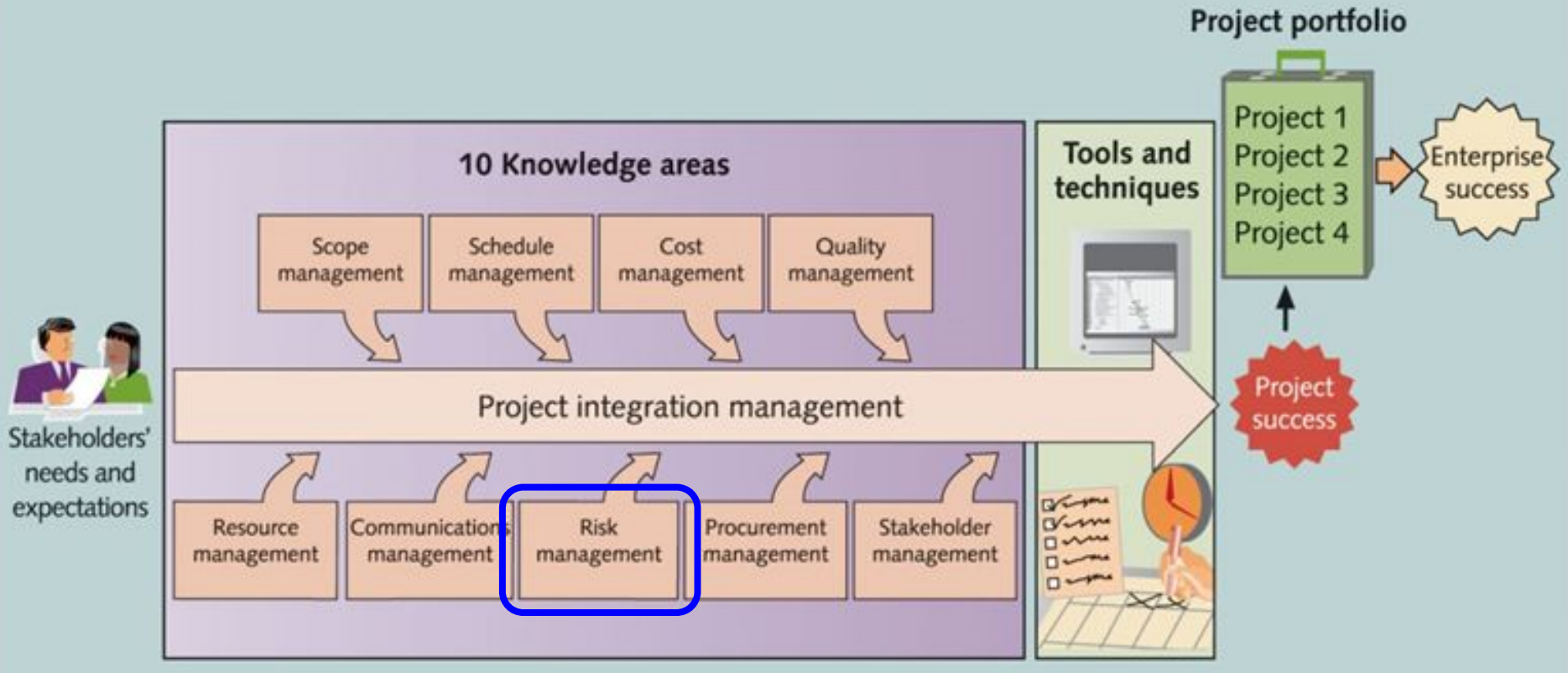
Lecture 7: Project **Risk** Management

—Identifying, Analyzing, and Mitigating Uncertainties—

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Project Risk Management



About this lecture



Learning objectives:

- Understand the **importance** of risk management
- **Identify and analyze** project risks
- Develop **risk response plans**
- **Control and monitor** project risks effectively

Introduction to Risk Management

What is Risk?

- A risk is an **uncertain event or condition** that, if it occurs, can have a **positive or negative effect** on project objectives
- **Every project has risks** that must be managed to ensure success

Example: Risks in a Mobile Banking App Project

1. **Scope Creep**

Continuous addition of new features without time or budget adjustments

Impact: Delays, cost overruns, and frustrated stakeholders

2. **Data Security Breach**

Sensitive customer data could be exposed due to poor encryption or authentication

Impact: Legal issues, loss of customer trust, brand damage

3. **Integration Issues**

Difficulty integrating with existing banking systems or third-party APIs

Impact: Functionality failure, increased development time

4. **Unclear Requirements**

Miscommunication between business analysts and developers

Impact: Rework, project delays

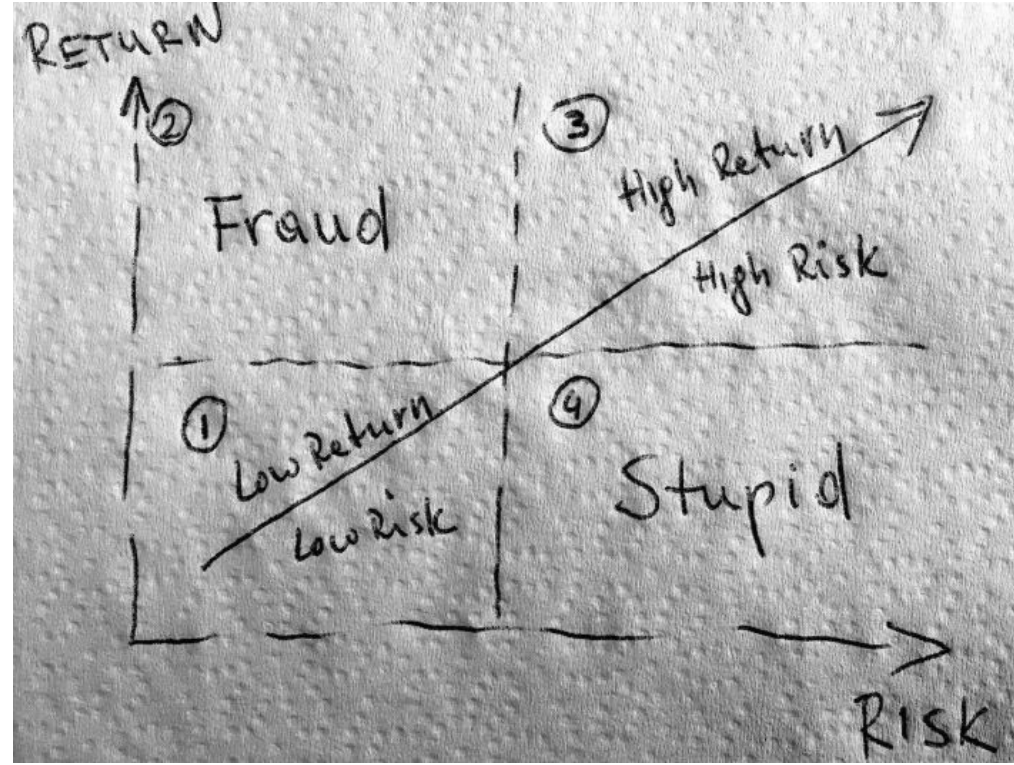
5. **Key Team Member Resignation**

Loss of technical knowledge and project momentum

Impact: Knowledge gaps, recruitment delays

Risk–Reward Relationship

- **Low risk – Low return**
 - Be careful of stagnancy
- **High risk – high return**
 - Be careful not to gamble
- **Low risk – High return**
 - Too good to be true
- **High risk – Low return**
 - Why would you do it??



The Importance of Risk Management

- **Uncertainty exists in all projects**
 - Risk management reduces surprises
- **Prevention is cheaper than correction**
 - Proactively handling risks saves costs
- **Higher risk, higher reward**
 - Calculated risks lead to innovation and success

Risk Management Process

The 4 Key Steps of Risk Management Process

1. **Risk Identification:** Listing potential risks
2. **Risk Analysis:** Assessing likelihood and impact
3. **Risk Response Planning:** Creating strategies to handle risks
4. **Risk Monitoring & Control:** Tracking and adjusting strategies

Step 1: Risk Identification

Why Risk Identification?

- Risk identification is the **first step** in risk management
- If risks are not identified early, they cannot be managed:
Unidentified Risk → Project Issue → Overruns & Delays
- Both **internal and external risks** should be considered

Common Risk Categories in IT Projects

1. Market Risks

2. Technical Risks

3. Financial Risks

4. Human Resource Risks

Common Risk Categories in IT Projects

1. Market Risk Examples

- Low customer demand for the system
- Aggressive competitor response or innovation
- Frequent changes in business priorities

Common Risk Categories in IT Projects

2. Technical Risk Examples

- Software bugs, integration failures, or server crashes
- New or untested technologies
- Security vulnerabilities (data breaches, cyber-attacks)

Common Risk Categories in IT Projects

3. Financial Risk Examples

- Budget overruns due to underestimation or scope creep
- Delayed funding or unexpected cost increases
- Vendor cost changes or contract issues

Common Risk Categories in IT Projects

4. Human Resource Risk Examples

- Key team members leaving during critical phases
- Lack of skilled personnel for specialized tasks
- Poor communication with stakeholders or within the team

Four Techniques for Identifying Risks

1. **Brainstorming:**

Group discussions to list possible risks

2. **Expert Consultation (Delphi Method):**

Anonymous surveys from experts to gather insights

3. **Project Document Review:**

Examining past project reports, lessons learned

4. **SWOT Analysis:**

Identifying Strengths, Weaknesses, Opportunities, and Threats

Identifying Risks: **Exercise**

*Imagine you're managing an IT project to **develop a new payment system for an online store**. What risks might you face?*

Identifying Risks: Example

Technical Risks

- **Integration Issues:** Difficulty integrating with third-party payment gateways (e.g., PayPal)
- **System Downtime:** Risk of system crashes during peak transaction times
- **Security Vulnerabilities:** Susceptibility to hacking, data breaches, or PCI non-compliance
- **Performance Problems:** Payment process may be slow or unreliable under load

Identifying Risks: Example

Project Management Risks

- **Scope Creep:** Uncontrolled expansion of features (e.g., adding support for cryptocurrency)
- **Unrealistic Deadlines:** Pressure to release too soon without proper testing
- **Inaccurate Estimates:** Underestimating time/cost required for secure and scalable solution

Identifying Risks: Example

Human Resource Risks

- **Lack of Expertise:** Team may lack experience in fintech or secure coding
- **Key Staff Turnover:** Loss of critical developers or project leads during development

Identifying Risks: Example

Compliance & Legal Risks

- **Regulatory Changes:** Sudden changes in financial regulations or tax rules
- **Data Protection Issues:** Violating GDPR or other privacy laws with customer payment data

Identifying Risks: Example

Testing & Quality Risks

- **Insufficient Testing:** Bugs in handling failed payments, currency conversion, or refunds
- **Edge Cases Not Covered:** Not testing rare but critical cases like chargebacks or fraud detection

Identifying Risks: Example

Stakeholder Risks

- **Misaligned Expectations:** Business owners may expect features out of the original scope
- **Poor Communication:** Delays caused by unclear requirements or feedback cycles

Identifying Risks: Example

External Risks

- **Third-Party Failures:** Downtime or errors from external services (e.g., banking APIs)
- **Market Competition:** Competitor launches a more advanced payment solution mid-project

Step 2: Risk Analysis

Why Risk Analysis?

- After identifying risks, we must **analyze their impact** to prioritize them
- Not all risks are equally dangerous:
 - Some risks are minor, while others can derail the entire project
- Risk analysis helps **decision-making** on **which risks need mitigation**

Two Types of Risk Analysis

- 1. Qualitative Analysis**
- 2. Quantitative Analysis**

Two Types of Risk Analysis: **Qualitative Analysis**

Qualitative Analysis:

- Uses **expert judgment** to **rank risks** based on likelihood & impact
→ Quick, subjective ranking

Example: Project delays due to unclear requirements

Two Types of Risk Analysis: **Quantitative Analysis**

Quantitative Analysis:

- Uses **mathematical models** to **estimate** financial or time impact
→ Data-driven, precise calculations

Example: Budget overrun estimation

Qualitative Risk Analysis: Risk Matrix

What is a Risk Matrix?

- A tool used to **prioritize risks** based on their **Likelihood** and **Impact**
 - Assign Likelihood: 1 (low), 2 (medium), 3 (high)
 - Assign Impact: 1 (low), 2 (medium), 3 (high)→ **Priority = Likelihood x Impact**

	Low Impact	Medium Impact	High Impact
High Likelihood	Medium	Critical	Critical
Medium Likelihood	Low	Medium	Critical
Low Likelihood	Low	Low	Medium

Qualitative Risk Analysis: Risk Matrix

How to Use the Matrix?

- **Focus on higher Likelihood, higher Impact risks first**
- Plan and respond to them, e.g., assigning budget reserves

	Low Impact	Medium Impact	High Impact
High Likelihood	Medium	Critical	Critical
Medium Likelihood	Low	Medium	Critical
Low Likelihood	Low	Low	Medium

Risk Matrix: Exercise

The Vinfast plant in Vietnam has identified the following risks. Assign Likelihood and Impact, then create a Risk Matrix.

- Minor price fluctuation
- Delay in material delivery
- Supply chain breakdown
- Small regulatory change
- Moderate cost overrun
- Inflation affecting project costs
- Slight labor cost increase
- Exchange rate changes
- Severe natural disaster

Risk Matrix: Example

	Low Impact	Medium Impact	High Impact
High Likelihood	Minor price fluctuation	Delay in material delivery	Supply chain breakdown
Medium Likelihood	Small regulatory change	Moderate cost overrun	Inflation affecting project costs
Low Likelihood	Slight labor cost increase	Exchange rate changes	Severe natural disaster

Quantitative Risk Analysis: Expected Monetary Value (EMV)

$$EMV = (Probability) \times (Impact)$$

- EMV helps assign a **monetary value** to risks
- Used for **financial risk assessment & insurance planning**

Quantitative Risk Analysis: Expected Monetary Value (EMV)

$$EMV = (Probability) \times (Impact)$$

Example:

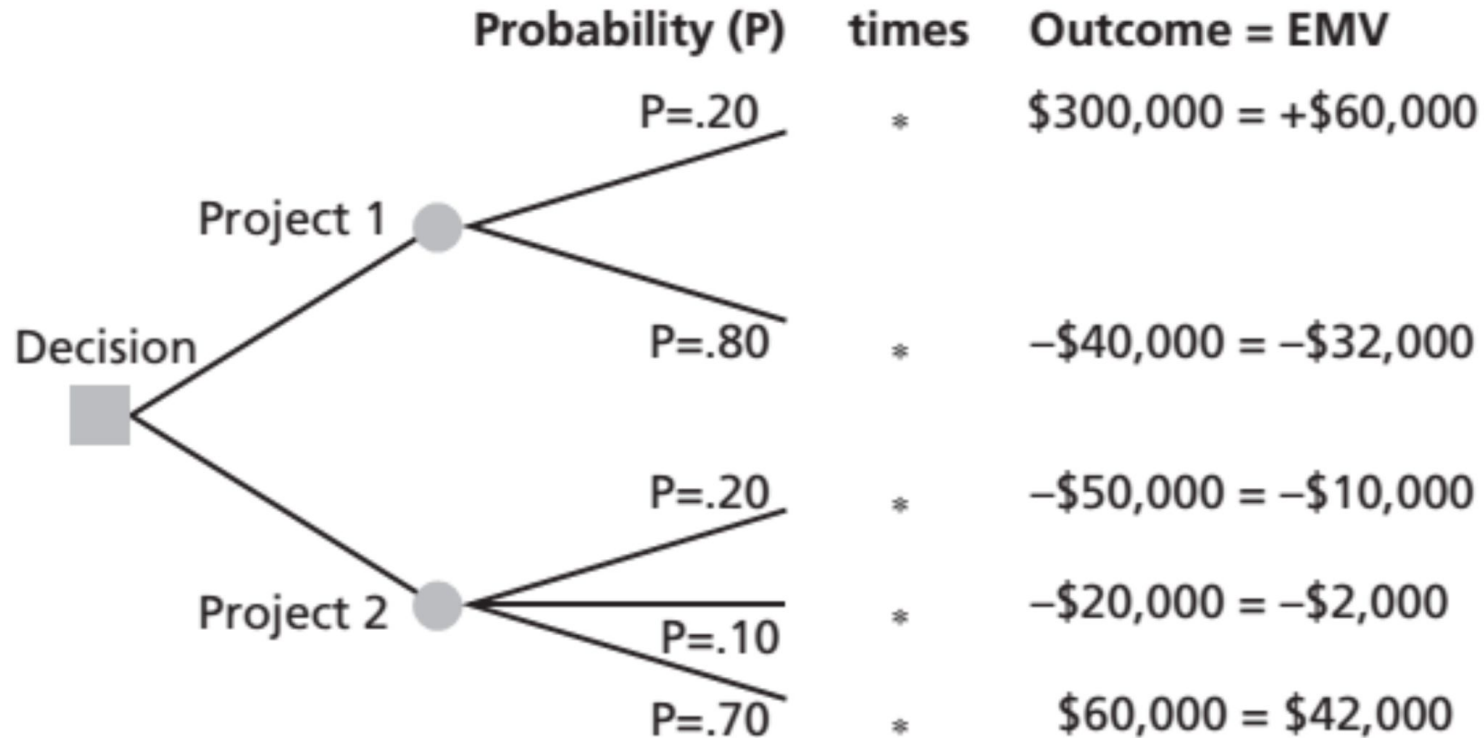
- **Event:** A potential delay in a project due to a supplier issue.
- **Probability:** 30% (0.3)
- **Impact:** A cost overrun of \$10,000

→ **EMV:** $0.3 \times \$10,000 = \$3,000$

Making Risk-Based Decisions: Decision Tree

- Visual tool to compare different **risk response options**
- Used to make **investment and strategic decisions**

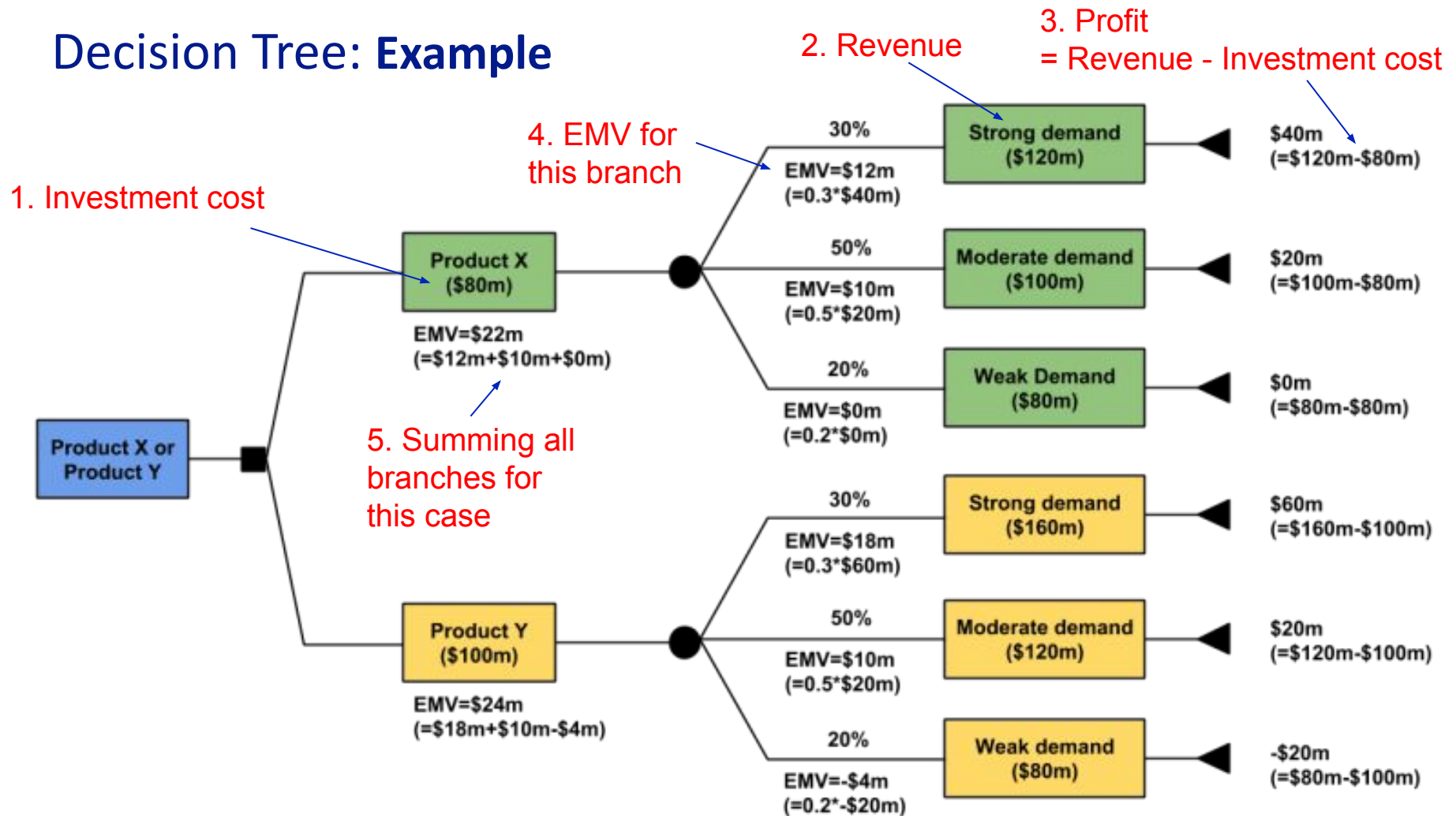
Decision Tree: Example



Project 1's EMV = $\$60,000 - \$32,000 = \$28,000$

Project 2's EMV = $-\$10,000 - \$2,000 + \$42,000 = \$30,000$

Decision Tree: Example



Step 3: Risk Response Strategies

Why Risk Response?

- **Risk response strategies** help **manage uncertainty** and **minimize negative effects**
- The **real challenge** in risk management is deciding **how to respond**
 - Depends on:
 - Probability of risk occurring
 - Impact on project goals
 - Cost of response vs. cost of risk

Four Risk Response Strategies

- 1. Risk Avoidance**
- 2. Risk Mitigation**
- 3. Risk Transfer**
- 4. Risk Acceptance**

Strategy 1: Risk Avoidance

Risk Avoidance: Change the project plan to remove the risk

- When to Use:
 - If the risk is **too dangerous or costly** to manage
 - If the project can be **restructured** to remove the risk entirely
- Example:
 - Instead of developing a new payment system from scratch (high risk), a bank can **use an existing, reliable payment gateway**

Strategy 2: Risk Mitigation

Risk Mitigation: Reduce the probability or impact of the risk

- When to Use:
 - When total avoidance is **impossible or impractical**
 - If a risk can be **reduced** with extra safety measures
- Example:
 - A software team worries about cybersecurity threats, they mitigate that by conducting **extra pen testing** and **encrypting data**

Strategy 3: Risk Transfer

Risk Transfer: Shift the risk to another party

- When to Use:
 - When the **financial impact** of a risk is too high to handle alone
 - If a **third party** can manage the risk better
- Example:
 - A company buys **insurance** to cover potential hacking damages
 - A bank **outsources** cybersecurity to a professional company

Strategy 4: Risk Acceptance

Risk Acceptance: Acknowledge the risk and plan contingencies (backup)

- When to Use:
 - When the risk is **low probability or low impact**
 - If the cost of other responses is **higher than the risk** itself
 - When there's **no way to prevent the risk**
- Example:
 - A startup launches a mobile app with a **small chance** of low initial downloads, and prepare contingency marketing plans
 - A website **acknowledges** downtime risk, and prepare backup servers

Risk Response Strategies: **Exercise**

A **banking system** faces cybersecurity threats. What should they do?

Risk Response Strategies: **Example**

1. **Avoid:** Do not build an online banking system (→ impractical)
2. **Mitigate:** Use multi-factor authentication and pen testing
3. **Transfer:** Buy cybersecurity insurance and hire professionals
4. **Accept:** Have a response plan in case of a breach

What is the best approach?

Can multiple strategies be used at once?

Responding to Positive Risks

Positive Risks: Turning Risks into Opportunities

- **Not all risks are bad!** Some risks are **opportunities**
→ Positive risks **must be managed strategically** to bring benefits

Four Strategies for Positive Risks

- 1. Exploiting Opportunities**
- 2. Enhancing Opportunities**
- 3. Sharing Opportunities**
- 4. Accepting Opportunities**

Strategy 1: Exploiting Opportunities

Exploiting Opportunities: Ensure the opportunity happens by taking proactive steps

When to Use:

- When the opportunity is **valuable and certain**
- When proactive action can **guarantee success**

Example:

- A software company finds a **faster development method**
→ Instead of just recognizing it, they **invest in the new method** and train all teams

Strategy 2: Enhancing Opportunities

Enhancing Opportunities: Increase the probability or impact of the opportunity

When to Use:

- When the opportunity is **uncertain but promising**
- When **small changes** can **increase likelihood or impact**

Example:

- A project is going very well and is ahead of schedule
→ **Enhancement:** Offer team incentives to **finish even earlier**

Strategy 3: Sharing Opportunities

Sharing Opportunities: Partnering with a third party to maximize the opportunity

When to Use:

- When another group **has the resources** to help maximize the opportunity
- When both sides **benefit from collaboration**

Example:

- A tech startup develops a **new AI algorithm** but lacks funding
→ **Sharing:** Partnering with a larger company to **commercialize the technology**

Strategy 4: Accepting Opportunities

Accepting Opportunities: Acknowledging and letting opportunities happen naturally

When to Use:

- When the opportunity **requires no additional effort**
- When the cost of enhancement or exploitation **outweighs** the benefit

Example:

- A construction project **under budget** due to material discounts
→ The team **accepts the savings** without making extra changes

Positive Risk Response Strategies: Exercise

A software company releases an app, and it becomes **more popular than expected**.

How should they respond?

Positive Risk Response Strategies: **Example**

1. **Exploit:** Invest in marketing to boost downloads even further
2. **Enhance:** Add new features quickly to retain users
3. **Share:** Partner with influencers and start monetizing
4. **Accept:** Enjoy the success without making changes

What is the best approach?

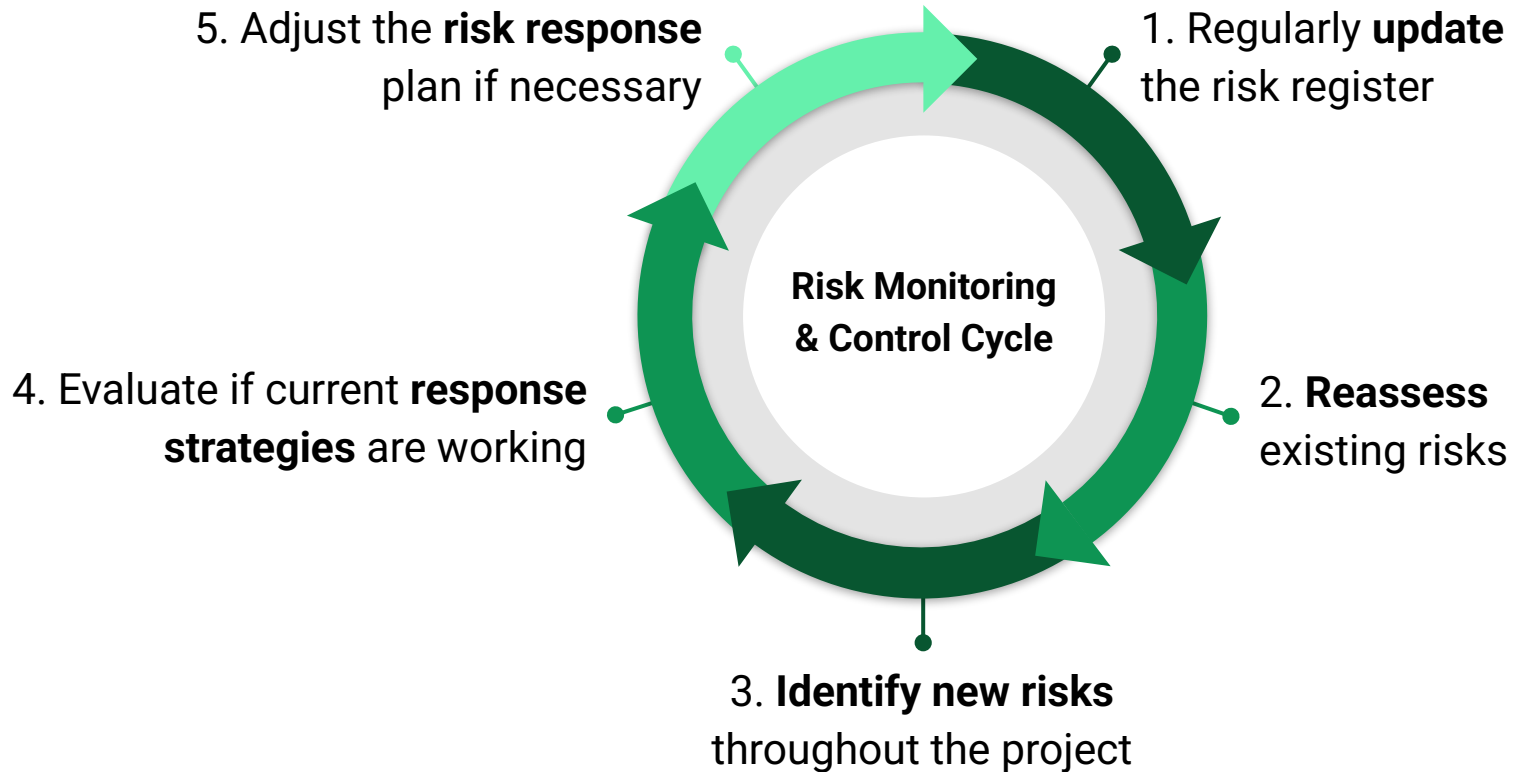
Can multiple strategies be used at once?

Step 4: Risk Monitoring & Control

Why Risk Monitoring & Control?

- Managing risks is **not a one-time task**, it's **continuous**
- New risks **emerge**, and old risks **change in impact and likelihood**
- **Failing to monitor risks** can lead to **unexpected project failures**

Risk Monitoring & Control Cycle



Approaches to Control Changing Risks

- **Escalate risks** (increase priority) if they become more severe
- **De-escalate risks** (decrease priority) if they are no longer a threat
- **Modify mitigation plans** if they are not effective
- **Develop new responses** for emerging risks

Example:

- A software company faces a data breach risk. Their **original plan was to conduct security tests every month**. After detecting **a rise in hacking attempts**, they **increase testing frequency to every week**.

Exercise

Risk Analysis: **Exercise 1**

- Draw Decision Tree and Calculate EMV
- Which project has lowest risk?

	Investment cost	Good case probability	Good case revenue	Bad case probability	Bad case revenue
Project 1	100000	30%	500000	70%	50000
Project 2	120000	20%	600000	80%	30000
Project 3	150000	25%	1000000	75%	20000

Summary

Summary



You have learned to:

- Understand the **importance** of risk management
- **Identify and analyze** project risks
- Develop **risk response plans**
- **Control and monitor** project risks effectively

Thank you for listening

Q & A

Appendix

Case Study:

Learning from Successes & Failures

- **Success Story:** NASA's Apollo Risk Management
- **Failure Case:** NASA's Challenger Disaster

Success Story: NASA's Apollo Risk Management

Problem: Space missions involve **extreme risks** (technical failures, life-threatening situations)

Risk Management Strategy:

- **Thorough risk analysis** before launch
- **Redundant safety systems** in case of failures
- **Simulated risk scenarios** to train astronauts

Outcome: Apollo 11 successfully landed on the Moon **despite risks**

Success Story: NASA's Apollo Risk Management

Example risk management:

- **Apollo 13 (1970):** When an oxygen tank exploded, the team **quickly adapted** using contingency plans
- Risk monitoring and **team problem-solving** saved lives

Lesson Learned: Proper risk planning and monitoring prevent disasters

Failure Case: NASA's Challenger Disaster (1986)

- **Problem:**
 - Engineers **warned about cold temperatures** affecting the **O-ring seals**
 - NASA knew about the issue, but ignored the risk
- **Outcome:** The Challenger exploded **73 seconds after launch**, killing **7 astronauts**

Failure Case: NASA's Challenger Disaster

What Went Wrong?

- Risk was **identified** but **not addressed**
- **Pressure to launch** led to ignoring safety concerns
- Poor **communication** between engineers and decision-makers

Success vs. Failure

Apollo 11	Challenger
Risks were identified early.	Risks were known but ignored.
Simulated failures before launch.	No proper risk contingency plan.
Safety measures were implemented.	Safety concerns were dismissed.
Adaptive response saved Apollo 13.	Rigid/slow decisions led to failure.

Lesson Learned:

- Identifying risks is not enough, you must also **act** on them

RISK MANAGEMENT FLOWCHART

Flowchart with Project Planning

